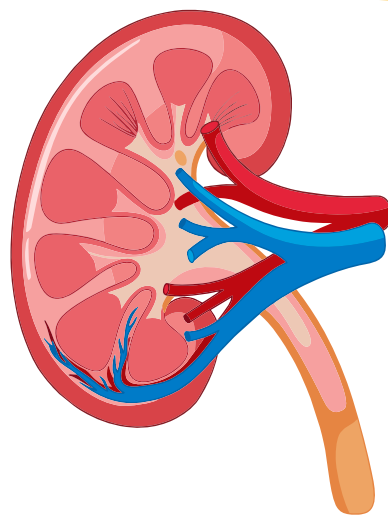
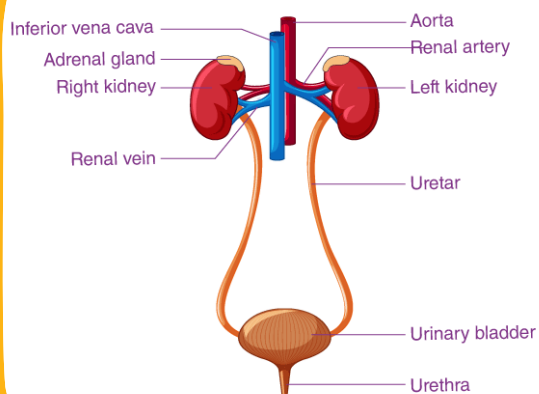




HOMEOSTASIS



HOMEOSTASIS

Homeostasis is the body's fundamental process of maintaining a stable and balanced internal environment despite external fluctuations. It involves a series of physiological mechanisms that regulate variables such as temperature, blood pressure, and glucose levels within narrow limits. The goal is to ensure optimal conditions for the body's cells and organs to function efficiently, contributing to overall health and well-being.

ADAPTATION OF PLANT FOR DIFFERENT INTERNAL CONDITION

Plants exhibit remarkable adaptations to thrive in diverse internal conditions, enabling them to flourish in various environments. There are three main aspect of homeostasis.

- **Osmoregulation:** Osmoregulation is the physiological process that regulates the balance of water and solutes in an organism's body to maintain internal stability.
- **Thermoregulation:** Thermoregulation is the biological process that helps maintain a stable internal body temperature in response to external temperature changes.
- **Excretion:** Excretion is the biological process of eliminating metabolic waste products and excess substances from an organism's body to maintain internal balance.

Excretion or storage of CO_2 :

Excretion or storage of CO_2 is crucial for regulating internal conditions. In animals, CO_2 is produced during respiration and removed through the lungs or stored in bicarbonate ions in the blood. In plants, CO_2 is absorbed during photosynthesis, converted into carbohydrates, and may be stored as starch or released through stomata when needed.

Removal of extra water:

Excess water in plants is removed primarily through transpiration, where water vapor escapes through stomata during photosynthesis, and occasionally through guttation, where excess water is exuded as droplets at leaf tips or margins due to root pressure.

- **Transpiration** is a crucial process in plants as it helps transport essential nutrients and minerals from the roots to the upper parts of the plant, maintaining its structure and facilitating photosynthesis. It also plays a role in cooling the plant and maintaining its water balance.

- **Guttation** on the other hand, occurs under specific conditions when the root pressure is high and the rate of transpiration is low, usually at night or in the early morning. It results in the formation of droplets at the leaf margins or tips. While not as common as transpiration, guttation can provide some relief from excess water uptake and pressure within the plant.

Osmotic adjustment in plants:

Osmotic adjustment in plants involves adjusting solute concentrations to maintain water balance in saline conditions.

Types of plants based on osmotic adjustment:

Hydrophytes:

- Adapted to thrive in aquatic or waterlogged environments.
- Features include floating leaves, well-developed air channels, and reduced cuticle to enhance gas exchange in water.

Halophytes:

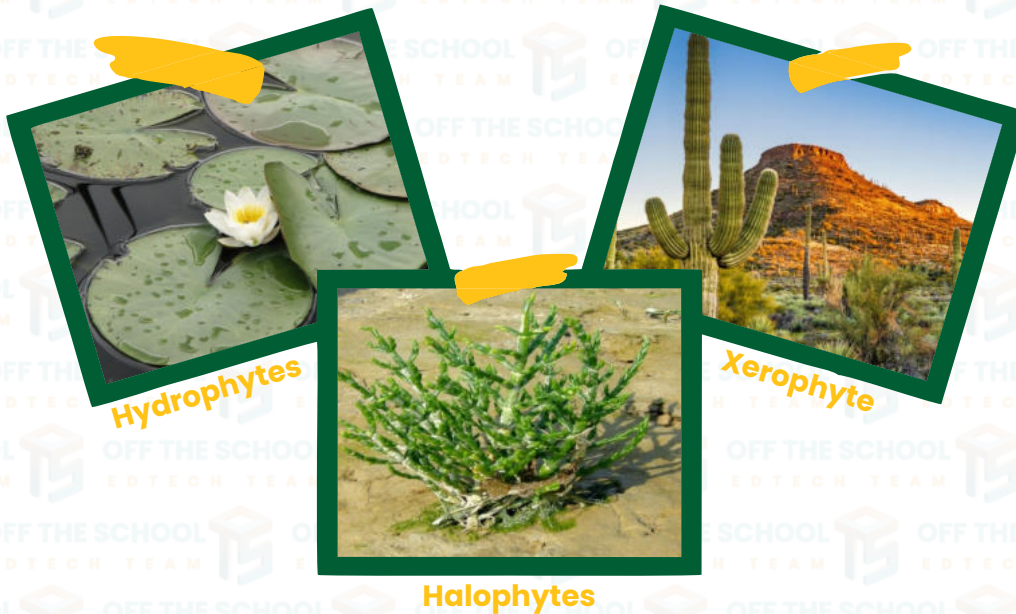
- Specially adapted to tolerate saline or salty conditions, such as coastal areas.
- Have mechanisms to excrete excess salt or sequester it in specialized cells to maintain osmotic balance.

Epiphytes:

- Grow on the surfaces of other plants, often trees, without being parasitic.
- Adaptations include aerial roots for support and absorption of nutrients and water from the surrounding environment.

Mesophytes:

- Adapted to moderate or average environmental conditions.
- Do not have specialized adaptations for extremely wet or dry conditions.
- Xerophytes:**
 - Specialized adaptations to survive in arid or water-scarce environments.
 - Features include reduced leaf size, water storage tissues, and mechanisms to minimize water loss.



HOMEOSTASIS IN ANIMALS

Osmoregulation In Animals:

Osmoregulation in animals is the physiological process of regulating the concentration of water and solutes in the body's internal environment to maintain proper hydration and internal balance, typically involving processes like filtration and secretion in the kidneys.

Osmoregulation in aquatic environment:

Osmoregulation in aquatic environments refers to the mechanisms used by aquatic organisms to regulate their internal water and solute balance in relation to the surrounding water.

Osmoregulation in fresh water:

Osmoregulation in fresh water in animals involves regulating internal salt concentrations to prevent excess water uptake from the hypoosmotic environment. Here are two differences between unicellular and multicellular animals:

CHARACTERISTIC	UNICELLULAR ANIMALS	MULTICELLULAR ANIMALS
Structure	Composed of a single cell	Composed of multiple cells
Function	Perform all life functions within a single cell	Specialized cells perform specific functions with division of labor
Examples	Amoeba, Paramecium	Humans, Dogs, Fish, etc.

Osmoregulation in marine animals:

Osmoregulation in marine animals is the process of regulating internal salt and water balance to adapt to the high salt concentrations of the ocean environment.

BONY FISH	CARTILAGINOUS FISH	OSMOCONFORMER
Bony skeleton	Cartilaginous skeleton	Composed of multiple cells
Operculum covers gills	Spiracles for breathing	Simple gill structure
Actively regulate salt balance	Limited osmoregulation ability	Conform to external salinity

Osmoregulation in terrestrial condition:

Osmoregulation in terrestrial conditions refers to the physiological processes by which landdwelling organisms maintain the balance of water and solute concentrations in their bodies.

This adaptation involves:

- **Water Conservation:** Terrestrial animals conserve water through efficient kidneys and concentrated urine.
- **Salt Regulation:** They manage salt levels with specialized salt glands or excretory structures.
- **Behavioral Adaptations:** Terrestrial animals exhibit behaviors like seeking shade and burrowing to cope with their environment.

Excretion:

Excretion is the biological process by which waste products and excess substances, such as metabolic waste and toxic materials, are removed from an organism's body to maintain internal homeostasis and prevent harm to the organism. It is a vital function that helps eliminate nitrogenous wastes (like urea and ammonia), excess salts, and other harmful compounds from the body, primarily through organs like the kidneys in mammals and birds or specialized excretory structures in other organisms.

Excretion in animals:

Excretion in animals is the elimination of metabolic waste products and excess substances from the body through specialized organs or structures. It helps maintain internal homeostasis and prevent the buildup of harmful compounds.

CHARACTERISTIC	PLANARIA	COCKROACH	VERTEBRATES	EARTHWORM
Excretory Organ	Protonephridia (Flame Cells)	Malpighian Tubules	Kidneys	Nephridia (Metanephridia)
Excretory Compound	Ammonia	Uric Acid	Urea	Urea
Excretory Source	Cellular Metabolism	Dietary Purine Compounds	Protein Metabolism	Protein Metabolism

HOMEOSTASIS IN MAN

Homeostasis in humans involves the regulation of internal conditions to maintain stability. The three main organs involved are the skin, lungs, and kidneys.

Skin: The skin serves as the body's primary temperature regulator by controlling heat loss and sweat production.

Lungs: Lungs facilitate the exchange of oxygen and carbon dioxide, helping to maintain the body's acid-base balance and overall oxygen levels.

Kidneys: Kidneys filter waste products and excess substances from the blood, regulating electrolyte balance, blood pressure, and fluid levels in the body, crucial for maintaining homeostasis.

Structure Of Human Skin:

The human skin has three primary layers:

Epidermis: The outermost layer, primarily consisting of flat, scale-like cells called keratinocytes, which provide protection.

Dermis: The middle layer, containing blood vessels, hair follicles, sweat glands, and nerve endings. It provides structural support and nourishment to the skin.

Subcutaneous Tissue (Hypodermis): The innermost layer, composed of fat cells (adipose tissue) that insulate the body, cushion organs, and store energy. These layers work together to maintain various skin functions, including protection, temperature regulation, and sensation.

Role of Skin in Regulating Body Temperature:

Thermoregulation: The skin plays a crucial role in maintaining the body's core temperature. When the body overheats, sweat glands in the skin produce sweat, which evaporates and cools the body. Conversely, in cold conditions, the skin's blood vessels constrict to reduce heat loss.

Insulation: Subcutaneous fat in the skin acts as an insulating layer, helping to conserve body heat in cold weather and preventing excessive heat loss.

Vasodilation and Vasoconstriction: Blood vessels in the skin can dilate (widen) or constrict (narrow) to regulate the flow of blood near the body's surface. Dilation releases heat, while constriction conserves it.

Body Temperature Rise:

- **Sweating:** The skin increases sweat production, which evaporates and cools the body.
- **Vasodilation:** Blood vessels near the skin's surface dilate to release heat.
- **Hair Erection:** Tiny muscles at the base of hair follicles contract, causing hairs to stand on end and trap a layer of insulating air.

Cold Conditions and Decreasing Body Temperature:

Vasoconstriction: Blood vessels near the skin's surface constrict to reduce blood flow and heat loss. **Shivering:** Muscles under the skin contract rapidly, generating heat. **Metabolism Increase:** Cellular metabolic activity may increase to generate more heat.

Role of lung to keep CO₂ concentration low on certain level:

Lungs play a vital role in regulating the concentration of carbon dioxide (CO₂) in the body to maintain it at an optimal level. When we breathe, the lungs remove excess CO₂ produced during cellular respiration. Oxygen is inhaled, and CO₂ is expelled from the bloodstream into the air sacs of the lungs. Through a process called diffusion, CO₂ is then exhaled when we breathe out. To ensure a low and controlled CO₂ concentration, the respiratory rate and depth can be adjusted. If CO₂ levels rise due to increased physical activity or other factors, the brain's respiratory centers can signal the respiratory muscles to breathe more rapidly and deeply, facilitating the removal of excess CO₂. Conversely, during periods of rest or reduced CO₂ production, the respiratory rate and depth can decrease to prevent excessive loss of CO₂. This delicate balance helps maintain an appropriate pH level in the blood and ensures that vital bodily functions continue to run smoothly.

Role of kidney in controlling blood composition:

The kidneys play a crucial role in controlling blood composition by filtering waste products, excess ions, and water from the blood, maintaining electrolyte balance, and regulating blood pressure. They also contribute to the production of erythropoietin, a hormone that stimulates red blood cell production.

URINARY SYSTEM OF MAN

The urinary system of a man functions to eliminate waste products and regulate fluid and electrolyte balance in the body.

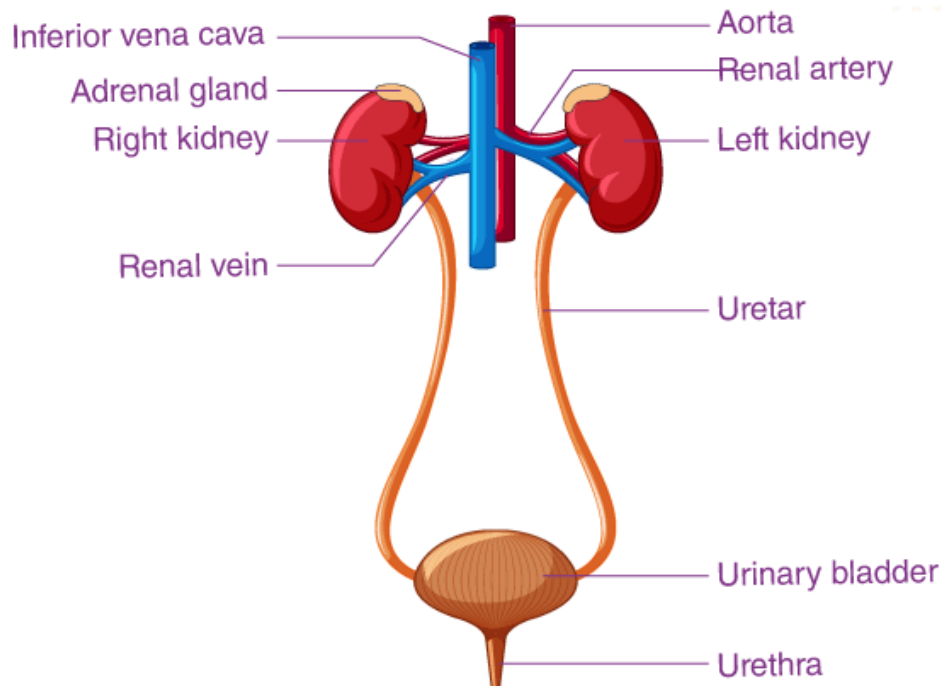
The urinary system of a man consists of:

Kidneys: Organs responsible for filtering waste products and excess substances from the blood to form urine.

Ureters: Tubes that transport urine from the kidneys to the bladder.

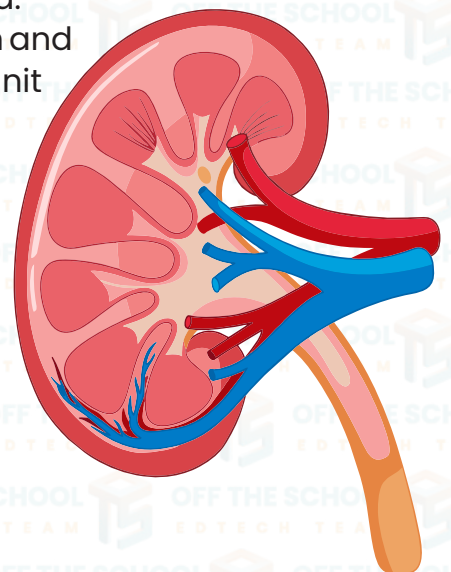
Bladder: A muscular organ that stores urine until it's ready to be eliminated from the body.

Urethra: A tube that carries urine from the bladder to the external environment during urination.



Structure of Kidney: The kidney is a bean-shaped organ located in the abdominal cavity. It consists of an outer renal cortex and an inner renal medulla.

Each kidney has a renal artery that brings blood to the organ and a renal vein that carries filtered blood away. The functional unit of the kidney is the nephron.



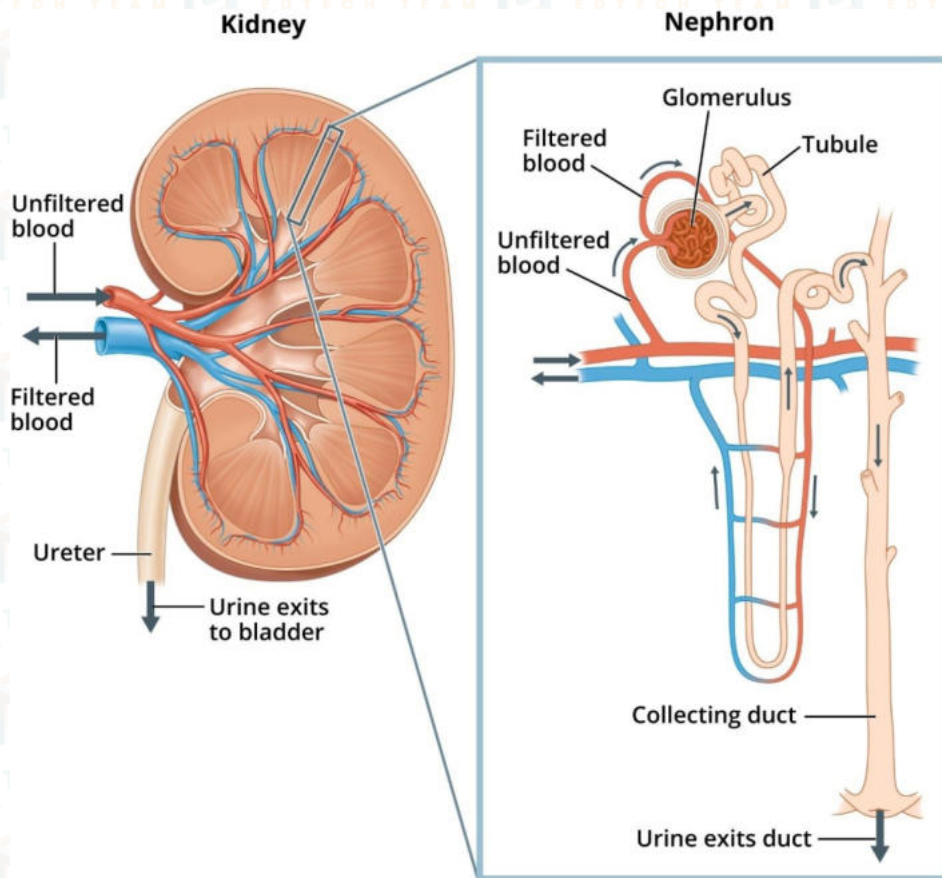
DID YOU KNOW?

Blood flow in your **kidneys** are higher than your **Heart, Liver & Brain!**



Structure of Nephron:

The nephron is the microscopic, functional unit of the kidney responsible for filtering blood and producing urine. Each nephron consists of a renal corpuscle and renal tubules. The renal corpuscle comprises the Bowman's capsule and glomerulus, where blood filtration begins. The renal tubules include the proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting duct. The nephron's intricate structure allows it to regulate the body's fluid balance, electrolytes, and waste removal through the formation of urine.



Function Of Nephrons:

Nephrons perform several crucial functions in maintaining the body's internal environment:

- **Filtration:** Nephrons filter the blood to remove waste products, excess ions, and toxins from the bloodstream. This initial filtration occurs in the glomerulus.
- **Reabsorption:** After filtration, nephrons selectively reabsorb essential substances, such as glucose, amino acids, and water, back into the bloodstream in the proximal convoluted tubule. This process ensures that valuable substances are not lost in the urine.
- **Secretion:** Nephrons also secrete certain substances, like hydrogen ions and potassium ions, into the tubules to maintain the body's acid-base balance and electrolyte levels.
- **Concentration of Urine:** Nephrons play a crucial role in adjusting the concentration of urine by reabsorbing or excreting water and solutes in the collecting duct. This helps regulate the body's fluid balance.

Role of kidney in urine formation:

Urea Formation: Urea formation, known as ureogenesis, is a metabolic process that occurs primarily in the liver. It involves the conversion of ammonia, a toxic waste product of protein metabolism, into urea, a less toxic compound, which is then excreted by the kidneys in the urine. This process allows the body to safely eliminate excess nitrogen while maintaining overall nitrogen balance.

Urine Formation:

Urine formation is the process of producing urine from blood plasma. The two main processes involved are **filtration**, where substances are filtered out of the blood, and reabsorption, where essential substances are selectively reabsorbed back into the blood while waste products are retained in the urine.

Filtration:

Filtration is the process of taking out materials from blood vessels into of two types: **Ultrafiltration**: Ultrafiltration is the process by which blood plasma is initially filtered into the renal tubules, allowing small molecules and ions to pass through the filtration barrier, including water, glucose, ions, and small molecules, while retaining larger molecules like proteins and blood cells in the blood vessels.

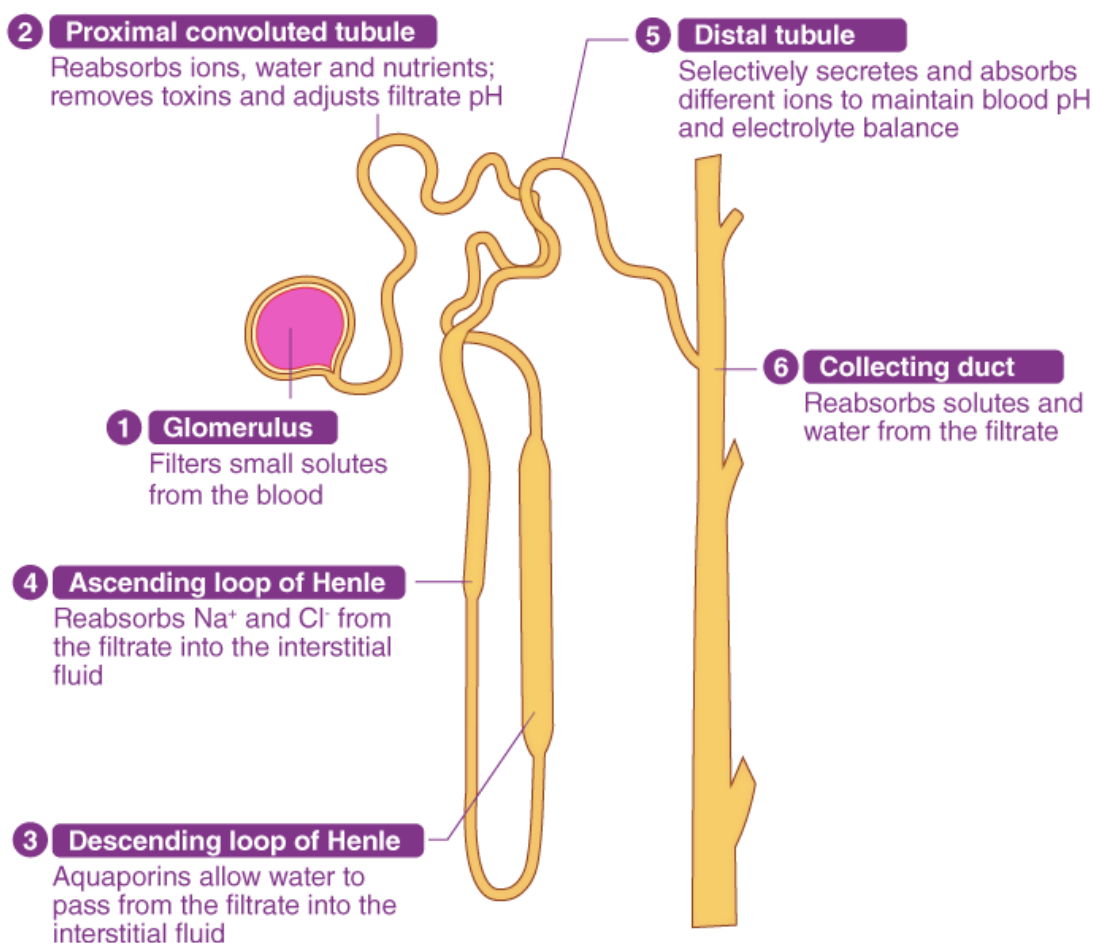
Selective Filtration: Selective filtration is the subsequent process in urine formation, where the renal tubules selectively reabsorb essential substances like glucose, ions, and water back into the bloodstream while actively secreting waste products and excess ions into the urine for excretion.

Reabsorption:

Reabsorption is the process in the renal tubules where essential substances like glucose, ions, and water are selectively transported back into the bloodstream, preventing their loss in urine.

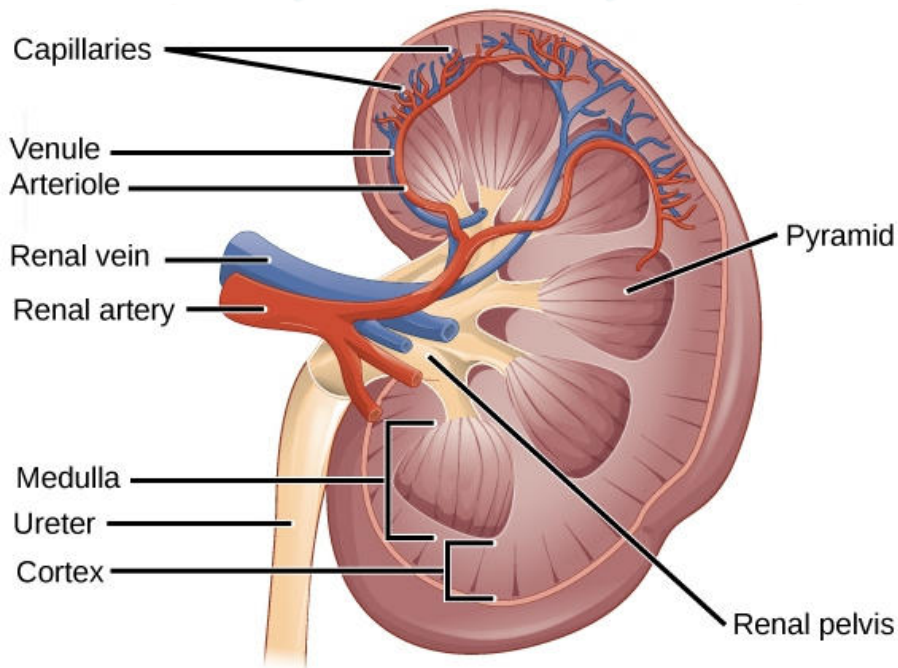
- **Non-Selective Reabsorption**: It is the passive reabsorption of water through osmosis, which occurs throughout the renal tubules.

- **Selective Reabsorption**: It is the active transport of specific substances (glucose, ions, etc.) from the renal tubules back into the bloodstream while allowing waste products to be excreted in urine.



Role of kidneys in osmoregulation:

The kidneys play a crucial role in osmoregulation, maintaining the body's water and solute balance. They filter blood to remove excess waste products, including ions and urea, while reabsorbing vital substances like glucose and ions. By regulating urine concentration and volume, the kidneys help stabilize blood osmolarity and maintain overall internal homeostasis.



DISORDERS OF KIDNEYS

Kidney Stones:

A kidney stone is a small, hard mineral deposit that forms in the kidneys when there is an accumulation of salts and minerals in the urine. These stones can vary in size and may cause severe pain when they obstruct the urinary tract. Treatment options include medication, dietary changes, or, in some cases, surgical removal.

Treatment Of Kidney Stones:

The treatment of kidney stones typically involves pain management with medication and drinking plenty of fluids to help pass the stone naturally. In some cases, larger stones may require medical procedures like lithotripsy or surgical removal.

Kidney Failure:

Kidney failure, also known as renal failure, is a condition in which the kidneys lose their ability to effectively filter waste and excess fluids from the blood. This can lead to the buildup of harmful substances in the body, resulting in various health complications. Dialysis or kidney transplantation is often required to manage advanced kidney failure.

Kidney Dialysis:

Kidney dialysis is a medical procedure used to artificially perform the filtration and purification functions of the kidneys when they are no longer able to do so effectively. During dialysis, a patient's blood is pumped through a machine that acts as an artificial kidney, removing waste products, excess fluids, and electrolytes. There are two main types of dialysis: hemodialysis, which uses an external machine, and peritoneal dialysis, which involves using the patient's abdominal lining as a filter. Dialysis is a life-saving treatment for individuals with kidney failure, but it requires regular sessions and strict adherence to dietary and fluid restrictions to maintain overall health.

Kidney Dialysis

